TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Real-time, in-flight droplet volume measurement for liquid handling (FastDrop)**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio·Medical		Electrical·Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Clinical Chemistry & Biological Analysis Instruments	Measuring Instruments	Measurement Sensors and Components
Concept of Proposed Technology	FastDrop is a compact, PCB-embedded capacitive sensing module that quantifies the volume of each droplet while it is in flight. Using miniaturized electrodes, lock-in detection, and advanced signal processing, it provides immediate, quantitative feedback without altering standard dispensing workflows. The system supports up to ~300 droplets per second and achieved an RMS error of ±5.15% across 50–360 nL in lab validation, with electrode geometries adaptable to other volume ranges.			
Rationale for Joint R&D Project	Korean SMEs developing automated liquid-handling tools, diagnostic instruments, and therapeutics manufacturing platforms increasingly operate in the sub-µL range, yet lack compact solutions for per-droplet volume verification. Gravimetric methods are offline and insensitive at these volumes, while optical systems are bulky and costly to integrate. A joint R&D project enables adaptation of FastDrop to partner-specific geometries, electronics, and software, and validation on the SME's dispensing platforms and liquids, accelerating time-to-product and de-risking commercialization for the Korean market.			
Key Words	1. In-flight droplet metrology 2. Capacitive sensing 3. Liquid handling QC 4. OEM integration 5. Sub-µL dispensing			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 7
R&D Objectives	Technical improvements at CSEM • Optimize electrode geometry and shielding to improve sensitivity and robustness across SME specific droplet volume ranges and pipette to plate distances. • Enhance on board detection electronics (replacing the external lock in amplifier) to achieve high SNR in a compact, OEM ready form factor. • Improve signal processing algorithms to compensate for droplet wobbling, mitigate XY/Z positional variability, and adapt to different classes of liquids. • Refine mechanical and electrical design of the sensing module, ensuring EMI/ESD robustness, manufacturability, and compatibility with integration constraints. Integration & joint development with SME • Mechanical and electrical integration of the FastDrop sensor into the SME's dispensing module (mounting, cabling, environmental shielding, sterilization/cleanability considerations). • Development of a basic control software and user interface enabling configuration, data visualization, and logging. • Pilot study and performance benchmarking on the SME's dispensing platform, including evaluation with real application liquids and workflow specific test sequences. • Validation against optical ground truth systems, definition of calibration and verification procedures, and establishment of reproducible measurement protocols.			
R&D Target Specifications	• Per-droplet measurement latency ≤ 5 ms (supporting ≥ 200–300 Hz dispensing). • Volume accuracy (RMS error) ≤ ±4% over target volume range defined with SME (baseline achieved ±5.15% over 50–360 nL). • Operational range (illustrative): 10–1000 nL with configurable electrode geometries; scalable beyond by design. • Compact sensor PCB to fit between pipette tip and plate (2–3 mm gap) with active shielding to mitigate tip interference. • On-board electronics option with SNR sufficient for single-droplet quantification without LIA; BOM optimized for OEM.			
R&D Plan	**WP1 — Requirements & Design Transfer (Month 0–1)** Objective: Joint definition of technical and integration requirements with the SME. Key tasks: • Collect SME-specific dispensing parameters (droplet volumes, dispensing speeds, mechanical clearances, fluid classes). • Define electrical, mechanical, EMC/ESD, and environmental constraints. • Define target performance metrics and acceptable tolerances. • Establish design-of-experiments (DoE) for later validation.			

Methodologies: Requirements engineering, interface specification, early feasibility checks.

Expected milestones:

- Finalized requirements document;
- Target volume range and mechanical constraints agreed.

WP2 — Sensor & Electronics Optimization (Month 1–4)

Objective: Improve sensing hardware to meet SME-specific performance needs.

Key tasks:

- Redesign electrode geometry and shielding to maximize SNR for the SME's volume range.
- Implement improved PCB layout with integrated shielding options.
- Develop and test on-board detection electronics to replace external LIA.
- Build first-iteration prototypes.

Methodologies: FEM simulations of electrode fields, PCB prototyping, benchtop electrical testing.

Expected milestones:

- Optimized electrode layout;
- On-board detection electronics v1;
- Prototype sensor PCB assembled.

WP3 — Signal Processing & Control Software (Month 2–6)

Objective: Increase robustness of volume extraction under real-world conditions.

Key tasks:

- Improve algorithms for wobble compensation and XY/Z positional variability.
- Adapt signal processing to multiple liquids (viscosity, conductivity).
- Implement basic control software/UI for sensor configuration and data visualization.

Methodologies: Algorithm refinement using laboratory datasets, controlled droplet-dispensing tests, software prototyping.

Expected milestones:

- Algorithm v2;
- Basic control UI;
- Demonstrated improvement vs. baseline accuracy.

WP4 — System Integration into SME Platform (Month 4–7)

Objective: Adapt the FastDrop sensor to the partner’s dispensing system.

Key tasks:

- Mechanical and electrical integration of the sensor PCB and electronics.
- Design fixtures, cable routing, shielding components, and environmental protection elements.
- Update firmware as needed for stable embedded operation.
- Perform first integrated bench tests on SME hardware.

Methodologies: Rapid prototyping, mechanical design iteration, electrical compatibility tests.

	Expected milestones: • Fully integrated sensor module on SME hardware. • Successful benchtop operation with real droplets. **WP5 — Validation & Calibration (Month 6–9)** Objective: Quantitative verification of performance vs. ground truth. Key tasks: • Perform side-by-side tests with optical reference system for droplet volume ground-truth (CSEM's DropWatch). • Test across different liquids, environmental conditions, and dispensing speeds. • Establish calibration curves and verification procedures. • Define measurement repeatability, accuracy, and robustness metrics. Methodologies: Optical metrology comparison, statistical analysis, calibration protocol development. Expected milestones: • Validated performance dataset. • Calibration/verification procedures ready for SME deployment. **WP6 — Pilot Study, Benchmarking & Transfer (Month 9–12)** Objective: Demonstrate the technology in real workflows and prepare for industrialization. Key tasks: • Conduct pilot trials on the SME's dispensing platform using application-specific workflows (e.g., assays, formulation steps). • Benchmark performance improvement vs. SME's baseline (error reduction, reproducibility, QC rate). • Identify reliability issues and finalize hardware/firmware improvements. • Provide documentation for integration, operation, and quality control. • Define next-steps for technology transfer and potential industrialization. Methodologies: Pilot process validation, workflow benchmarking, reliability testing. Expected milestones: • Pilot results report. • Validated TRL-7 performance. • Technology transfer package delivered

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead sensor/electronics design, simulations, and prototyping (PCB-based electrodes, active shielding, on-board detection). • Develop signal processing and calibration algorithms; provide control SW and data formats. • Set up validation with optical ground-truth and provide performance datasets (including raw data) and documentation. • Train SME engineers; joint test planning and execution; contribute to documentation for instrumentation. • Support IP strategy and technology transfer; co-author SOPs and verification procedures.
Expected R&R of Korean SME Partner	• Provide application requirements, target volumes, dispensing hardware and liquids; define performance acceptance criteria. • Co-develop mechanical/electrical interfaces; implement integration in host software; perform in-house tests. • Contribute to validation (bench and pilot); run comparative tests vs. existing QC; provide feedback for optimization. • Assess manufacturability, reliability, and service; plan regulatory pathway (as applicable) and market introduction. • Lead Korean-side coordination and contribute to joint dissemination, exploitation, and potential joint IP.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Validated, integrable droplet metrology module enabling per-droplet QC in sub-µL dispensing workflows. • Improved yield, reproducibility, and traceability for Korean SME platforms in Advanced Bio (gene/cell therapies, vaccines, diagnostics). • A validated hardware/firmware/software prototype (sensor PCB, on-board electronics, basic control software) with initial calibration guidelines. • Data packs and SOPs supporting reliability testing with potential customers (TRL7) and paving the way for industrialization. • Foundations for longer-term co-development, licensing, and potential joint IP and localization for Korean market needs.

6. References

(1) FastDrop: Capacitive sensor for in-flight droplet volume measurement, CSEM STR 2025 (to be published in Feb 2026).

(2) https://www.csem.ch/en/news/hamilton--csems-dropwatch-technology/

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization